

Website Category Classification for the Dexar Focus Assistant: From a TF-IDF Baseline to a Fine-Tuned ModernBERT

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Abstract

This project builds a deep-learning classifier that, given a URL, fetches the page, extracts the main textual content from its HTML, and predicts the website’s category over a 25-class taxonomy. The work follows an experimental narrative across three model families: a classical TF-IDF baseline with a linear classifier, a from-scratch PyTorch text model (a multi-kernel 1-D convolutional network and a bidirectional LSTM with attention pooling), and a fine-tuned `answerdotai/ModernBERT-base` encoder. On a held-out test set of 1,822 pages, the fine-tuned ModernBERT reaches a macro-F1 of 0.507 ± 0.004 and an accuracy of 0.561 ± 0.007 averaged over three seeds, ahead of the TF-IDF baseline (0.473 macro-F1) and the from-scratch model (0.319 macro-F1). The headline metric is macro-F1 rather than accuracy because the class distribution is imbalanced. The absolute scores are bounded by two factors that the analysis quantifies: a training corpus reduced to roughly 6,300 English pages after cleaning, and single-source, unverified labels that impose a label-noise ceiling. Ablations show that fine-tuning depth is the dominant hyperparameter, with a frozen encoder collapsing to a macro-F1 of 0.090 against 0.538 for full fine-tuning. The classifier is built as a component of Dexar, an on-device focus assistant, where a topical category for the active browser page supplies a signal for detecting cognitive drift.

Introduction and Task

The task is single-label multi-class classification over 25 categories, restricted to English-language websites. The input to each model is the main textual content extracted from a page’s HTML, not the raw markup and not the bare URL string. The practical objective is an inference path that accepts a live URL, retrieves and cleans the page, and returns the most probable categories with their associated probabilities.

This classifier is built as a component of Dexar, an on-device focus assistant for macOS that detects cognitive drift by comparing a user’s declared task against their observed activity. Dexar’s watcher reads active-window metadata, which includes the page URL when the foreground application is a browser, but a browser’s window title is often a generic or truncated string that carries little topical signal. Assigning the visited page to one of 25 topical categories provides a stable and interpretable signal of what kind of site the user is viewing, independent of an uninformative title. A declared task implies a set of expected categories, so a visited page that falls into a distractor category such as Gambling, Shopping, or Games is evidence of drift even when a title-based similarity check is inconclusive. This downstream use is the reason the input is page content rather than the bare URL string, and the reason the taxonomy retains fine-grained distractor categories.

The class distribution is imbalanced. In the raw dataset the largest category (Health) holds 1,093 pages against 131 for the smallest (Adult), and the same skew persists in the cleaned test set, where Health has a support of 165 and Adult a support of 20. Accuracy alone rewards a model that performs well on frequent classes while ignoring rare ones, so the headline metric throughout this report is macro-averaged F1, which weights every class equally. Accuracy and a per-class confusion matrix are reported alongside it for completeness. This choice follows the evidence in Homepage2Vec, where a macro-F1 of 0.90 on a balanced setup fell to 0.634 on the natural, unbalanced distribution, with the gap concentrated in the small classes (Lugeon et al., 2022).

Related Work

The project sits at the intersection of two studied problems: website and web-page classification, and the content-extraction step that classification depends on.

Website and web-page classification

Homepage2Vec is the closest prior work (*Epfl-Dlab/Homepage2vec*, 2026; Lugeon et al., 2022). It releases more than two million labelled sites from Curlie, the human-edited directory that succeeded DMOZ, under 14 top-level categories aligned across 92 languages, and trains a single network over textual, visual, domain-name, and HTML-metadata features. Two design choices carry over to this project. First, Homepage2Vec ships a library that fetches a URL and classifies it, which is the inference shape implemented here in `fetch.py`. Second, it handles imbalance by reweighting positive samples by the negative-to-positive ratio, which motivates the inverse-frequency weighted cross-entropy used for all trainable models here. The differences also set expectations: Homepage2Vec is multi-label and multilingual, whereas this project is single-label, English-only, and uses a finer 25-class taxonomy, so a lower F1 than its 14-class numbers is the anticipated outcome rather than a deficiency.

Nandanwar and Choudhary fine-tune BERT with a softmax head on the WebKB and DMOZ datasets, reaching F1 scores of 96 percent and 84 percent respectively (Nandanwar & Choudhary, 2023). Their work validates the transformer step and supplies two specific guides. They identify maximum sequence length as the hyperparameter that most affects performance, which motivates the 256, 512, and 1024 sequence-length ablation in this project. They report classical machine-learning and recurrent baselines (support vector machines, LSTM, and BiLSTM) below their fine-tuned BERT on both datasets, which provides literature support for the expected ordering from classical methods to transformers. This project deliberately diverges from their preprocessing: they stem and strip stopwords, which suits older embeddings but is counter-productive for a subword transformer, so ModernBERT here is fed raw extracted text. Classic BERT caps at 512 tokens, a constraint that ModernBERT’s native long context relaxes and that makes the 1024-token setting feasible.

URLBERT pre-trains a URL-string encoder on three billion URLs and classifies web pages from the URL alone (Li et al., 2025). It represents the URL-only alternative to this project’s content-based approach. Its own framing, that URLs are loosely structured and semantically sparse, is the justification for classifying fetched page content rather than the bare URL string. Sasazawa and Sogawa classify web pages with large language models for a different objective, distinguishing index pages from content pages to support crawlers (Sasazawa & Sogawa, 2025). Two points transfer: a zero-shot or few-shot language-model classifier is a low-cost option for further work, and their dataset is built by automatic annotation with a manual-review pass, the same weak-label risk that informed the data decisions below.

Content extraction and boilerplate removal

The HTML-to-text step is itself a studied task, so extraction quality is a genuine source of error rather than a free preprocessing stage. Web2Text combines convolutional potentials over 128 DOM-based block features with a hidden Markov model decoded by Viterbi, set a state-of-the-art on the CleanEval boilerplate-removal benchmark, and showed that better content extraction measurably improves downstream retrieval (Vogels et al., 2018). BoilerNet replaces hand-crafted features with a BiLSTM sequence labeller over raw HTML tags and words (Leonhardt et al., 2020). Its central finding, that feature-based extractors tuned on one era of web pages degrade on newer pages while a featureless model generalizes, is the reason this project relies on *trafilatura*, a modern and maintained extractor, rather than older heuristics, and treats residual extraction errors such as cookie and copyright notices that read as fluent text as a documented limitation.

Data

Dataset selection

The primary dataset is `massimilianowosz/website_categories`, which provides 15,186 raw training rows and 3,797 raw test rows, with raw HTML in the `text` column, URLs present, a predefined train/test split,

and 25 modern categories. The raw HTML makes content extraction a genuine pipeline step, and the presence of URLs enables the live-fetch inference path. Several alternative datasets were investigated and rejected, summarized in the inventory below.

Dataset	Decision	Reason
<code>massimilianowosz_website_categories</code>	Primary	25 modern categories, raw HTML, URLs present, predefined split.
<code>datablations/c4-filter</code>	Rejected as labels	1.66M rows but labels applied by an automatic classifier (weak labels) on a dated 2019 crawl.
Website Classification by Hetul Mehta	Rejected	Only 1,408 rows, a different 16-category taxonomy, text pre-cleaned by an unreproducible pipeline.
DMOZ	Rejected	Outdated directory, thus not of help for our application.

Training on automatically labelled data would introduce label noise as a design choice, which is the reason the C4-derived file was rejected despite its size.

Cleaning pipeline

The cleaning pipeline (`data.py`) concatenates the raw parquet shards and then applies a sequence of filters, each of which is counted and recorded in `dataset_report.json`. Main content is extracted with `trafilatura`, with a `BeautifulSoup` fallback (`extract.py`). Pages whose raw HTML matches a bot-wall, error-page, parked-domain, or JavaScript-shell pattern are flagged as garbage and dropped, as are extractions shorter than 200 characters. Remaining rows are deduplicated first by normalized URL, which lowercases the URL and strips the scheme, a leading `www.`, and any trailing slash, and then by a hash of normalized content to catch mirrored pages. Language is detected with `langdetect` on the first 2,000 characters of extracted text, and only English pages are kept. The pipeline then removes from training any URL that also appears in test, and asserts that the resulting train/test URL overlap is zero. The cleaned training rows are split 90/10 into training and validation by stratified sampling on the category label with a fixed seed of 42, while the official test split is left untouched as the held-out set. The funnel from raw to kept is shown below.

Stage	Train	Test
raw	15,186	3,797
minus garbage, failed, or short extraction	-3,512	-822
minus duplicate URL or content	-261	-23
minus non-English (<code>langdetect</code>)	-4,342	-1,130
minus train/test URL leakage	-35	n/a
kept	7,037	1,822

The kept training rows divide into 6,333 training and 704 validation pages.

Exploratory analysis

The exploratory analysis surfaces three properties that shape the modeling. The class distribution, shown in the figure below, confirms the roughly eleven-fold imbalance between the largest and smallest categories and is the reason macro-F1 is the headline metric.

The single most consequential finding is that only about 62 percent of the dataset is English. The English-only filter mandated by the project specification therefore removes roughly 38 percent of pages and is the largest

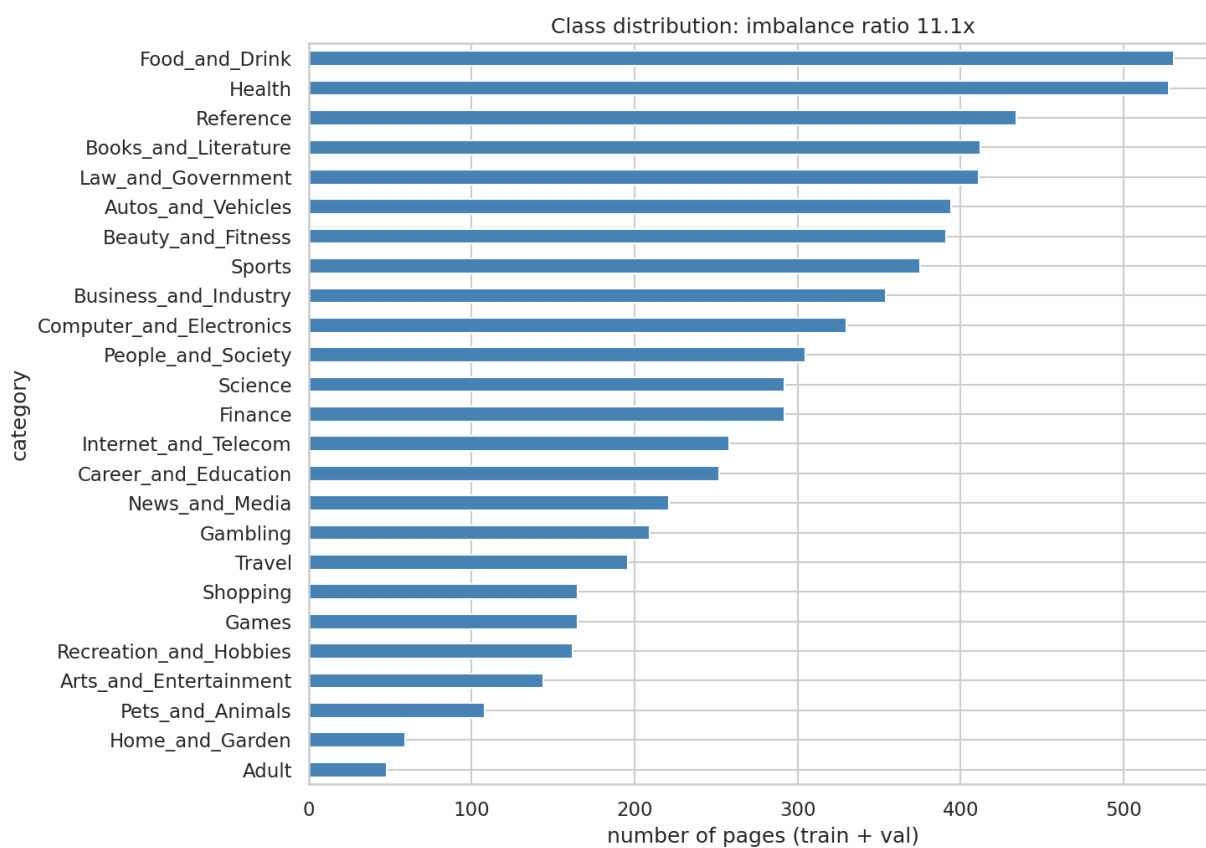


Figure 1: Per-class page counts in the cleaned dataset. The roughly eleven-fold gap between the largest and smallest categories motivates a macro-averaged headline metric.

single source of data loss, which reduces the usable training corpus to about 6,300 rows, well below the proposal’s earlier estimate of 16,000 to 17,000. This is a per-specification outcome, verified by inspecting langdetect output on samples, not a defect.

Content length before and after extraction quantifies the boilerplate-removal effect, comparing raw HTML length against extracted-text length.

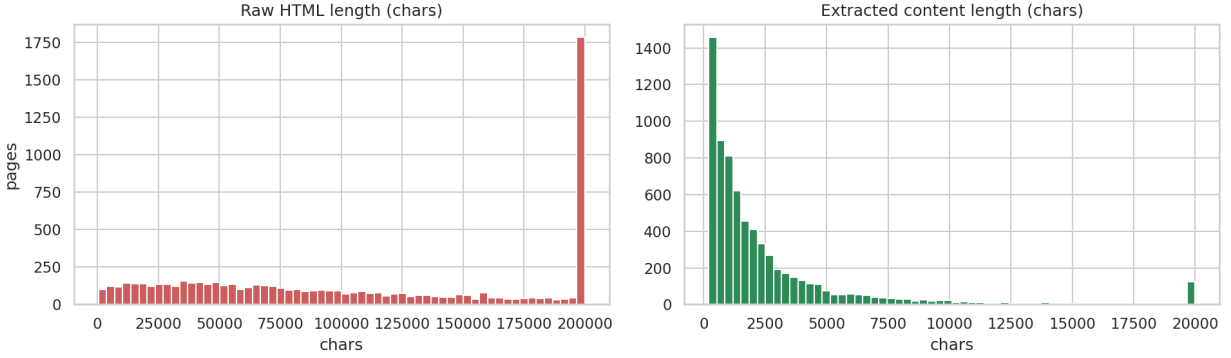


Figure 2: Raw HTML length against extracted-text length. Extraction removes navigation, scripts, and markup, reducing each page to its main content.

The token-length distribution under the ModernBERT tokenizer, with truncation lines at 256, 512, and 1024 tokens, justifies the sequence-length settings in the ablation and shows what fraction of pages each limit truncates.

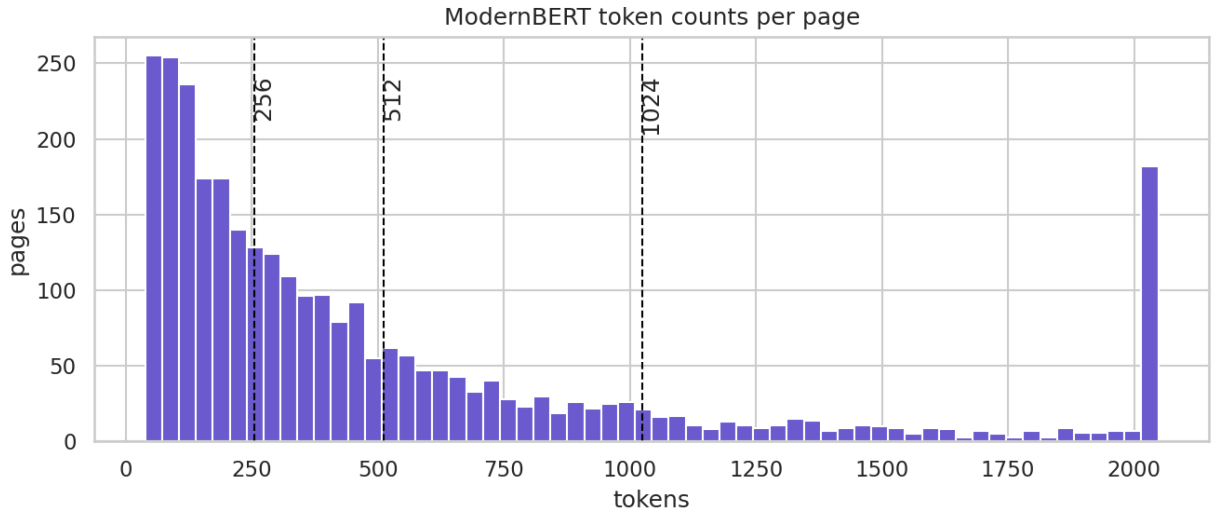


Figure 3: ModernBERT token counts per page with truncation thresholds marked at 256, 512, and 1024 tokens.

Top vocabulary per class, summarized as word clouds for four representative categories, gives a qualitative read on class separability and on the lexical overlap that drives later confusion between adjacent categories.

Approach

The end-to-end pipeline is shown below: a URL is fetched, its raw HTML is reduced to clean main content, and that content is tokenized and classified into top-k categories with probabilities. The same extraction and



Figure 4: Most frequent vocabulary for four representative classes, illustrating both distinctive terms and cross-class overlap.

tokenization stages are shared between training and live inference.

Content extraction

Content extraction (`extract.py`) runs `trafilatura` with settings that favor recall, retaining tables and removing comments and duplicated blocks. When `trafilatura` returns nothing or fewer than the minimum number of characters, a BeautifulSoup fallback strips scripts, styles, and similar non-content tags and returns the visible text, and the pipeline keeps whichever extractor produced more content. The minimum content threshold is 200 characters for training, while inference passes a smaller threshold so that thin but real pages still receive a best-effort prediction.

Classical baseline

The baseline is a non-deep reference point: TF-IDF features over word unigrams and bigrams feeding a linear support vector classifier (`LinearSVC`). It establishes the score that any deep model must exceed to justify its added complexity.

From-scratch model

The from-scratch family (`models.py`) provides two architectures, both trained from random initialization, which satisfies the requirement to build and adapt a model rather than reuse a pre-trained one. The convolutional classifier follows the Kim-2014 design: an embedding layer feeds parallel 1-D convolutions with kernel sizes 3, 4, and 5 and 128 filters each, followed by max-over-time pooling, concatenation, and a two-layer feed-forward head. The recurrent classifier embeds the input, runs a bidirectional LSTM, and pools the hidden states with additive attention that masks padded positions before a comparable head. Both share a hidden dimension of 256, while the embedding dimension and dropout are selected on the validation set from a small grid over embedding dimensions of 128 and 200 and dropouts of 0.3 and 0.5. The architecture with the higher validation macro-F1, the BiLSTM, is the one carried to the test set, with the grid selecting an embedding dimension of 128 and a dropout of 0.5. Tokenization uses a frequency-capped word vocabulary built from the training corpus, limited to 30,000 types with a minimum frequency of 2. Training uses AdamW at a learning rate of 1e-3, gradient-norm clipping at 5.0, inverse-frequency weighted cross-entropy, and early stopping on validation macro-F1 with a patience of 3.

Main model

The main model fine-tunes `answerdotai/ModernBERT-base`, a 2024 encoder with native long-context support, using the HuggingFace sequence-classification head (`train.py`). Inputs are subword-tokenized eagerly so that epochs are free of tokenization overhead. Optimization uses AdamW with decoupled weight decay, a linear warmup over the first 10 percent of steps followed by linear decay, `bfloat16` mixed-precision autocast on CUDA, gradient accumulation of two steps, and inverse-frequency weighted cross-entropy. The classifier pooling strategy, the maximum sequence length, the learning rate, the use of class weights, and the fine-tuning depth are all controlled by parameters and form the axes of the ablation study. Fine-tuning depth ranges over a full fine-tune, a last-N-layers variant that unfreezes the final transformer blocks, and a linear-probe variant that freezes the encoder and trains only the head. Training early-stops on validation macro-F1 with a patience of 2, and the best checkpoint is saved.

Inference pipeline

The inference path (`fetch.py`) fetches a URL over HTTP with a timeout and a user-agent header, extracts content with the same `trafilatura` pipeline used in training, tokenizes it, runs the fine-tuned ModernBERT, and returns the top-k categories with softmax probabilities. Pages that the garbage detector identifies as bot walls, error pages, or JavaScript-only shells are reported as an explicit error with a non-zero exit code rather than producing a prediction from non-content text.

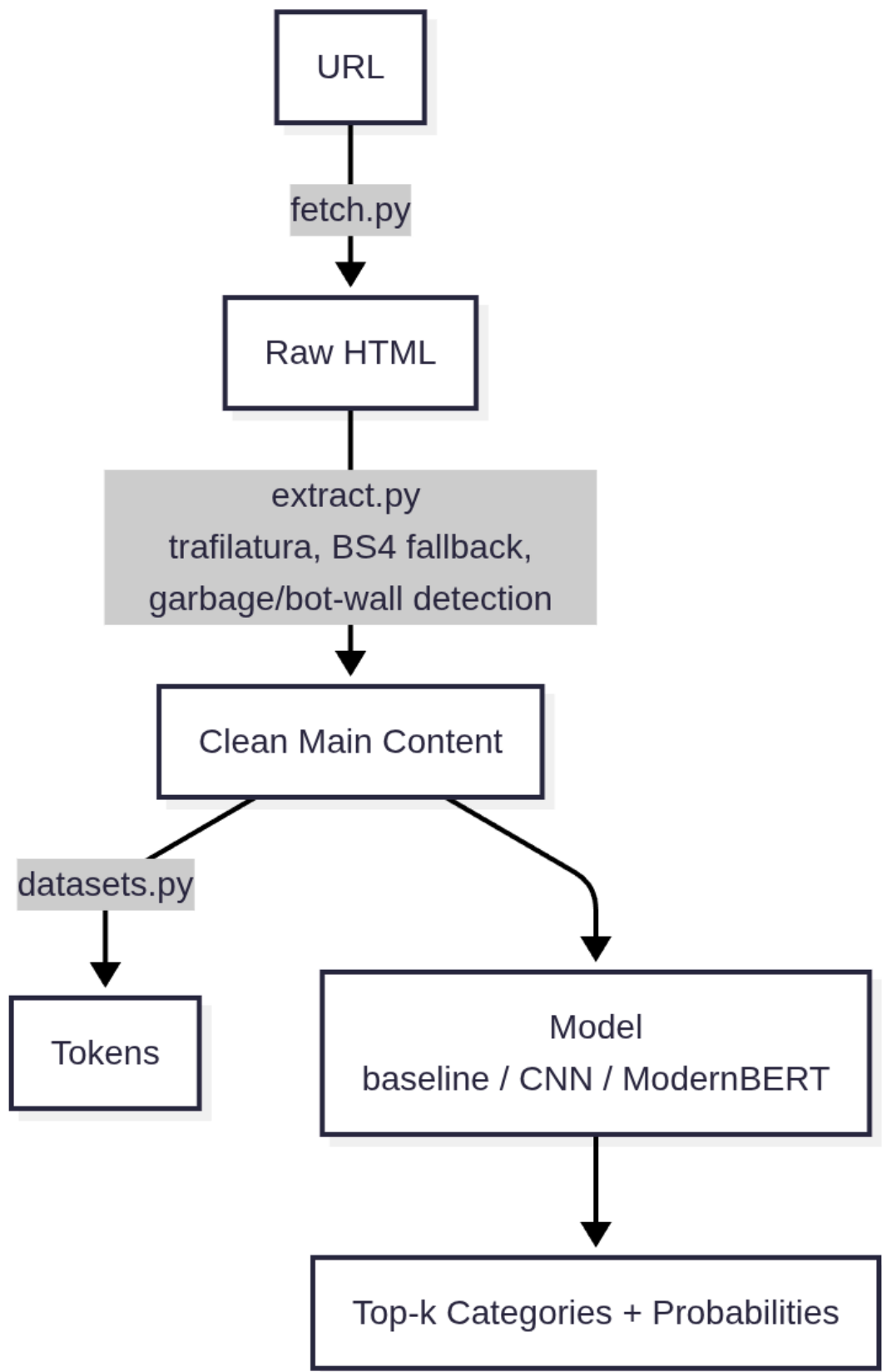


Figure 5: The classification pipeline from a URL to ranked category predictions, with the module responsible for each stage labelled.

Reproducibility

All random number generators are seeded, cuDNN is pinned to deterministic mode, and split indices are persisted. The test set is held out for final reporting only: every model-selection and hyperparameter decision is taken on the validation set, and the test set is never used for tuning. The final ModernBERT is trained under three seeds (42, 43, and 44), each evaluated once on the test set, and the headline figures are their mean and standard deviation. Bitwise-identical CUDA results are not enforced, because the BiLSTM has no deterministic cuDNN backward kernel and enabling strict determinism would raise at runtime; all reported figures are the values recorded in `artifacts/results/*.json`.

Results

On the held-out test set, the model ordering follows the proposal’s expectation: ModernBERT exceeds the classical baseline, which in turn exceeds the from-scratch model. The headline numbers are summarized below and visualized across the three families.

Model	Test macro-F1	Test accuracy
TF-IDF + LinearSVC (baseline)	0.473	0.533
From-scratch BiLSTM	0.319	0.364
ModernBERT (full fine-tune, 3-seed mean $\pm$ std)	0.507 ± 0.004	0.561 ± 0.007

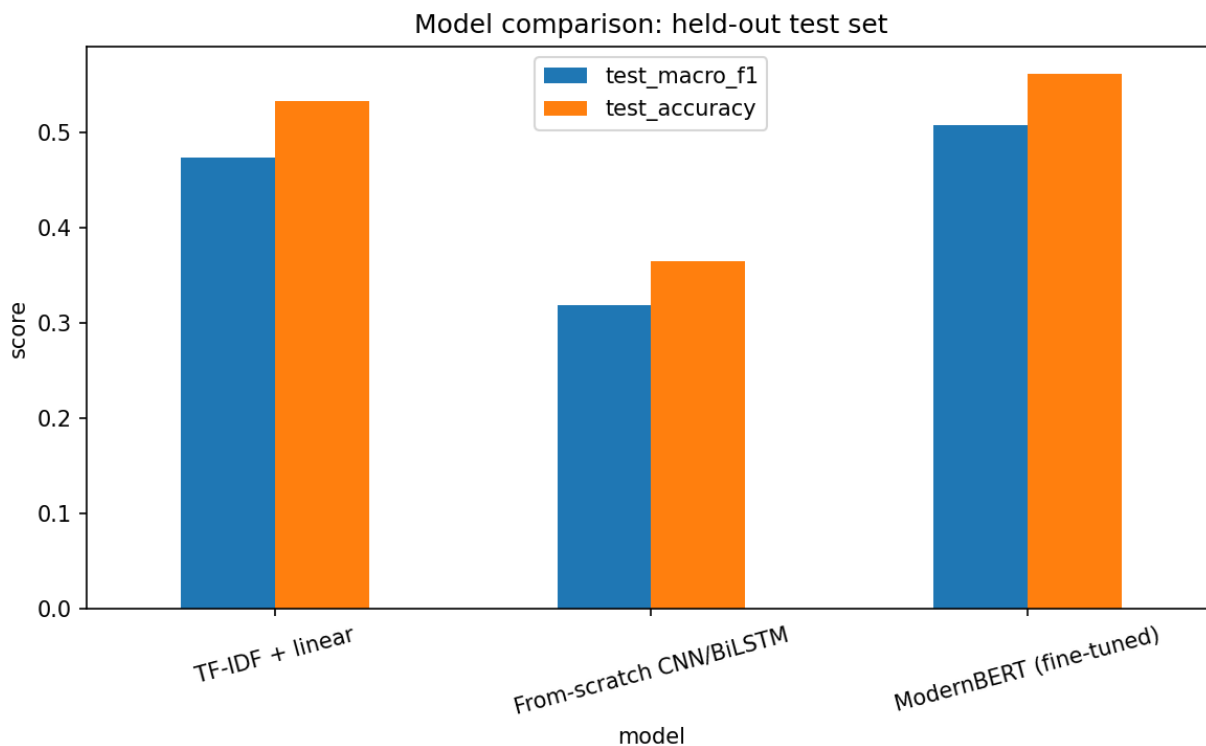


Figure 6: Test macro-F1 and accuracy across the three model families. ModernBERT leads, the TF-IDF baseline is second, and the from-scratch model trails both.

The from-scratch model trailing a well-tuned TF-IDF baseline, and ModernBERT exceeding that baseline by a narrow margin, are honest outcomes on a corpus of about 6,300 training rows rather than failed runs. The

proposal anticipated both: a from-scratch CNN or BiLSTM on this many documents need not beat a tuned linear model, and the transformer’s advantage is expected to be present but modest at this corpus size.

The ModernBERT training curve shows validation macro-F1 peaking at epoch 3 with a value of 0.534, after which the model overfits and early stopping selects the epoch-3 checkpoint.

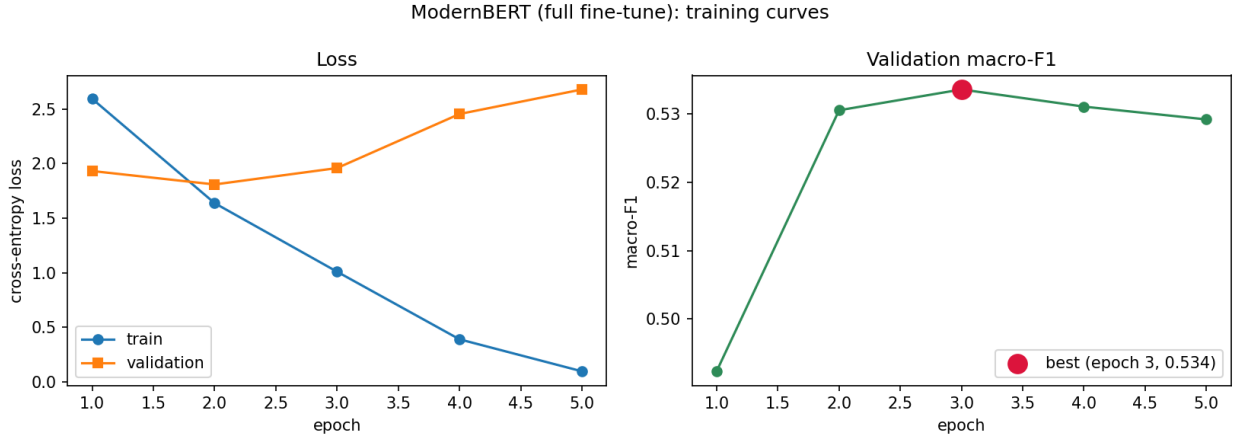


Figure 7: ModernBERT training and validation loss and validation macro-F1 per epoch. Validation macro-F1 peaks at epoch 3, after which later epochs overfit.

The row-normalized confusion matrices for the three models locate the errors. A strong diagonal indicates high per-class recall, and off-diagonal mass concentrates in semantically adjacent categories.

Per-class F1 for ModernBERT, sorted ascending with class support annotated, shows that performance tracks both support and category breadth. The strongest classes are Food_and_Drink at 0.74, Finance at 0.74, and Health at 0.72. The weakest are News_and_Media at 0.17 against a support of 63, Adult at 0.18, and Arts_and_Entertainment at 0.23 against a support of 40, which are either small classes or semantically broad categories that overlap many others.

Ablations

The ablation study sweeps each axis over a reduced two-epoch budget and reports validation macro-F1.

Axis	Settings and scores	Winner
Learning rate	1e-5: 0.454, 3e-5: 0.528, 5e-5: 0.555	5e-5
Sequence length	256: 0.519, 512: 0.538, 1024: 0.540	1024
Pooling	CLS: 0.540, mean: 0.535	CLS
Class weighting	on: 0.541, off: 0.528	on
Fine-tuning depth	head: 0.090, last-N: 0.377, full: 0.538	full

Fine-tuning depth is the dominant axis by a wide margin. A frozen encoder used as a linear probe collapses to 0.090, unfreezing the last layers recovers to 0.377, and full fine-tuning reaches 0.538, which establishes that adapting the encoder weights, not only the head, is what makes the transformer effective on this task. Among the remaining axes the effects are much smaller: a learning rate of 5e-5 is clearly best, while sequence length, pooling, and class weighting each move validation macro-F1 by at most about 0.02. A sequence length of 1024 edges out 512 by a marginal 0.002, consistent with the token-length distribution showing that a nontrivial share of pages exceed 512 tokens, which ModernBERT’s native long context can accommodate; CLS pooling and class weighting likewise give small gains.

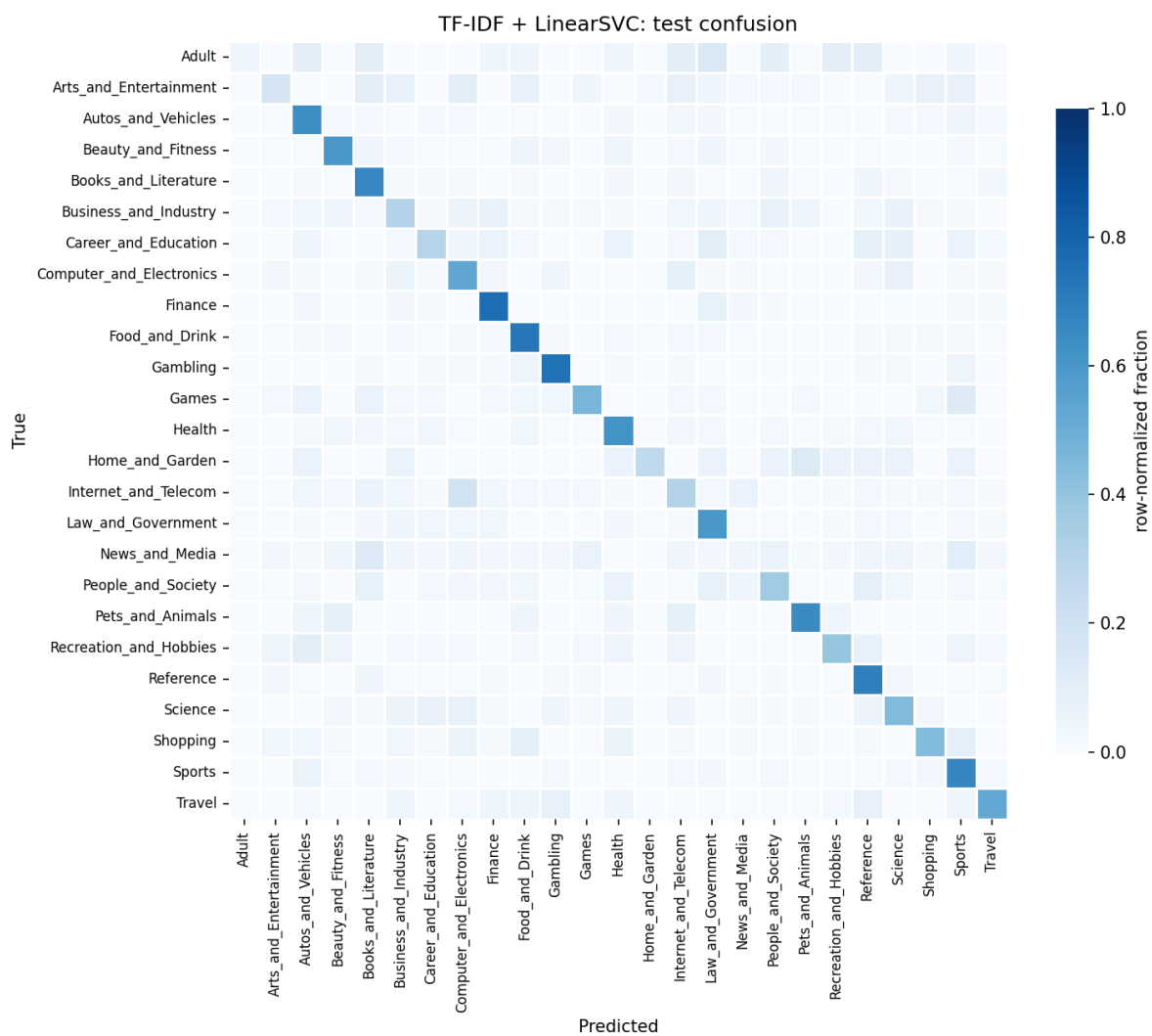


Figure 8: Row-normalized test confusion matrix for the TF-IDF baseline.

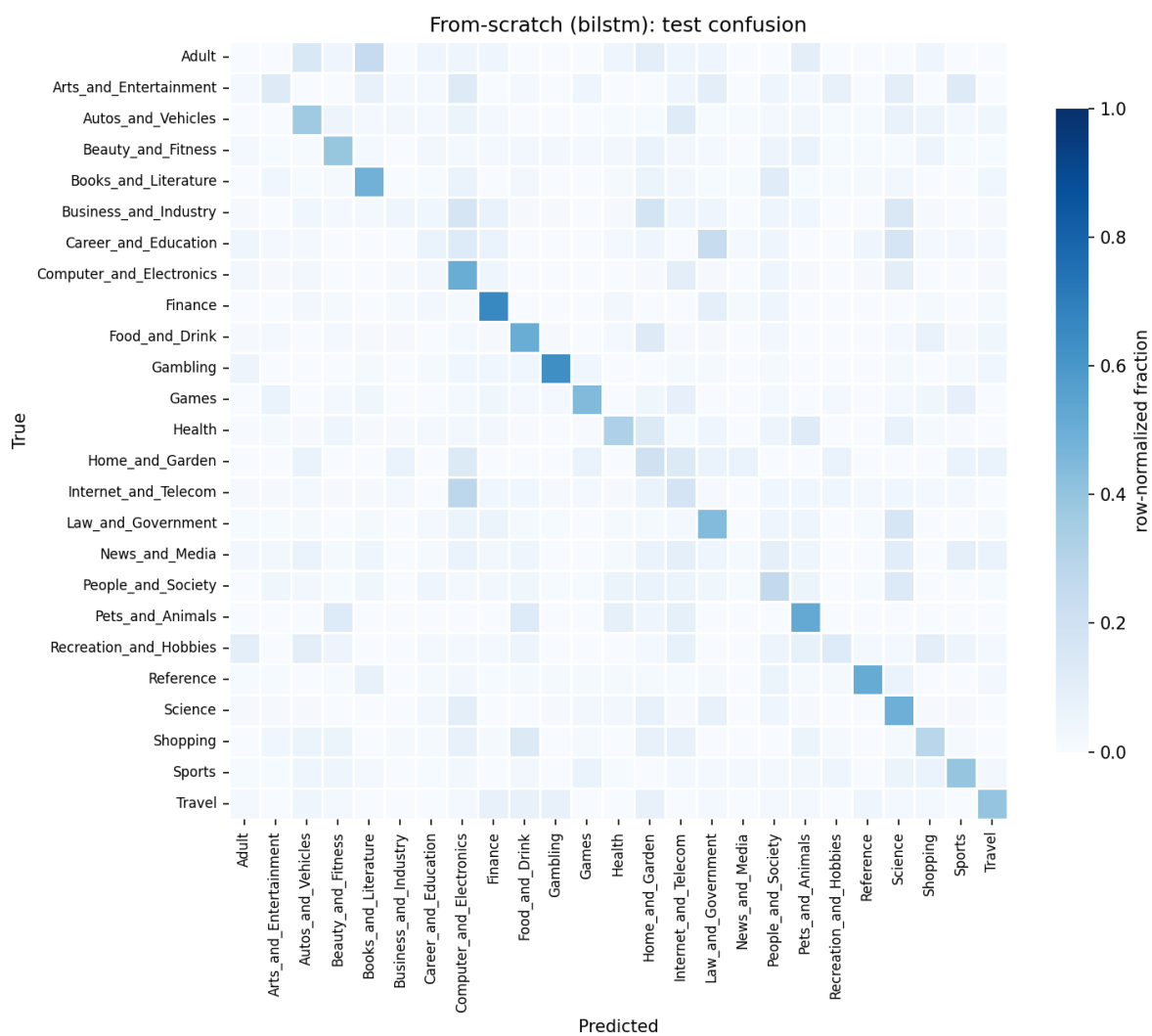


Figure 9: Row-normalized test confusion matrix for the from-scratch model.

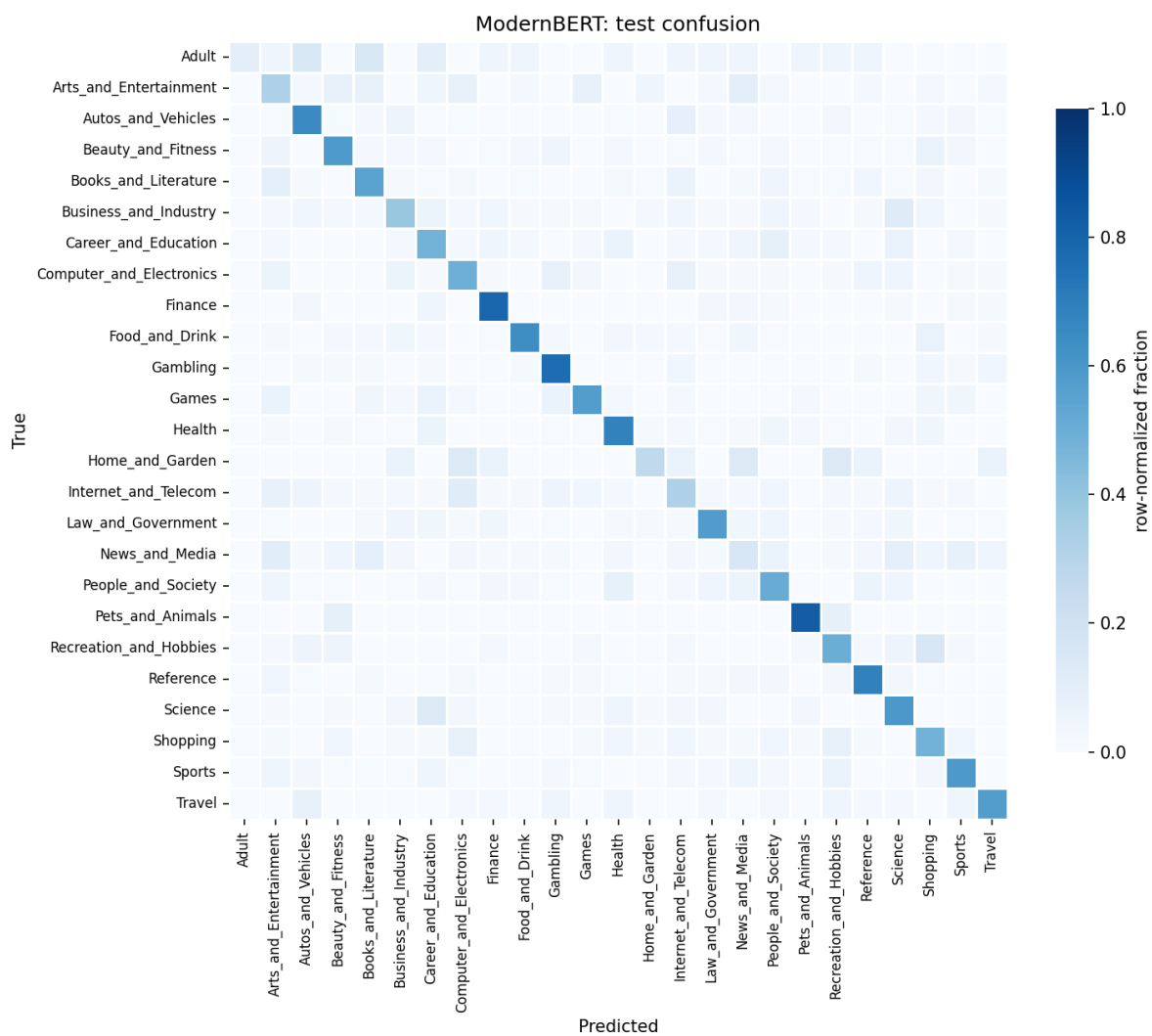


Figure 10: Row-normalized test confusion matrix for ModernBERT.

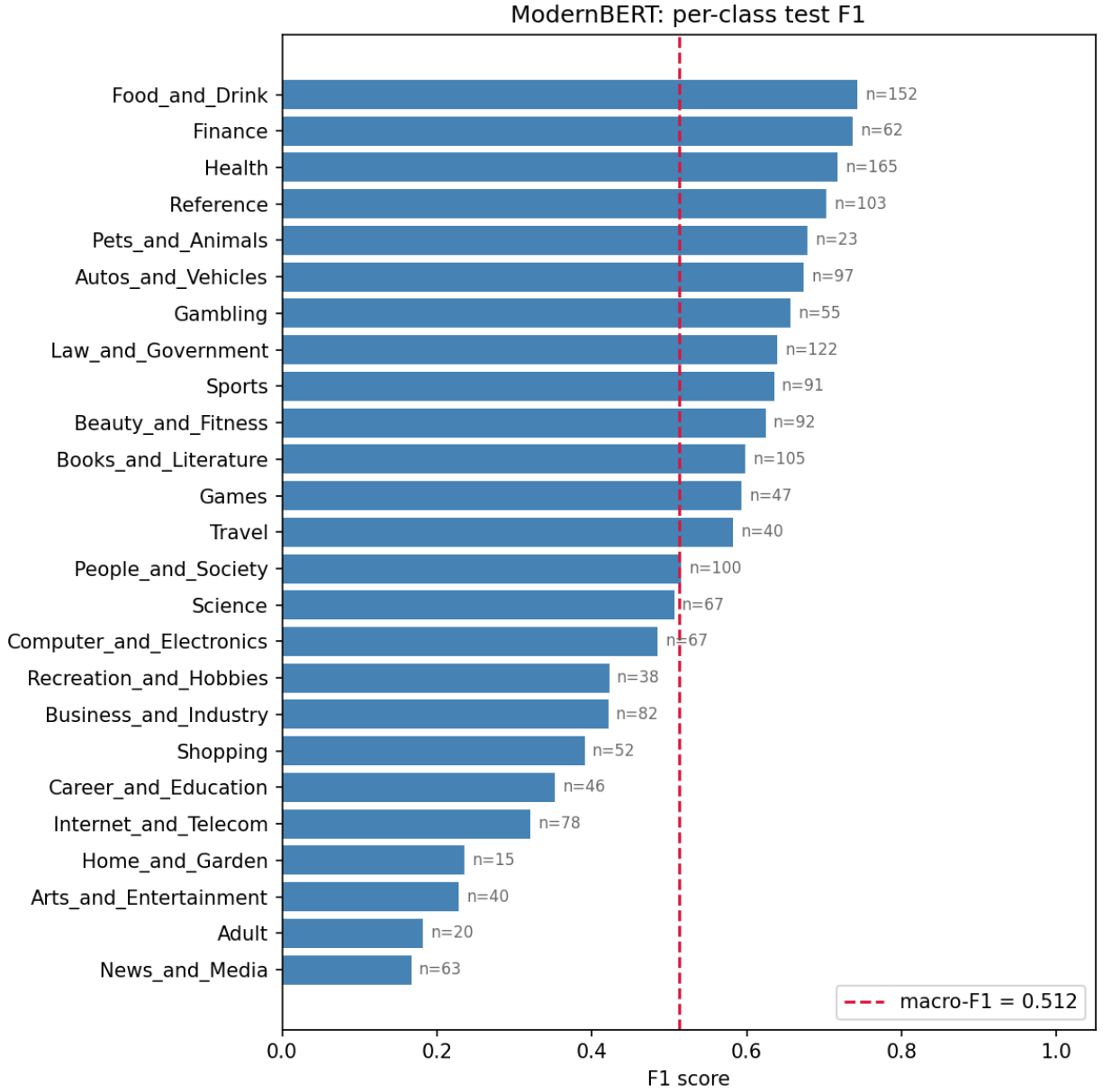


Figure 11: Per-class test F1 for ModernBERT, sorted ascending, with class support annotated and the macro-F1 line marked. Small and broad classes score lowest.

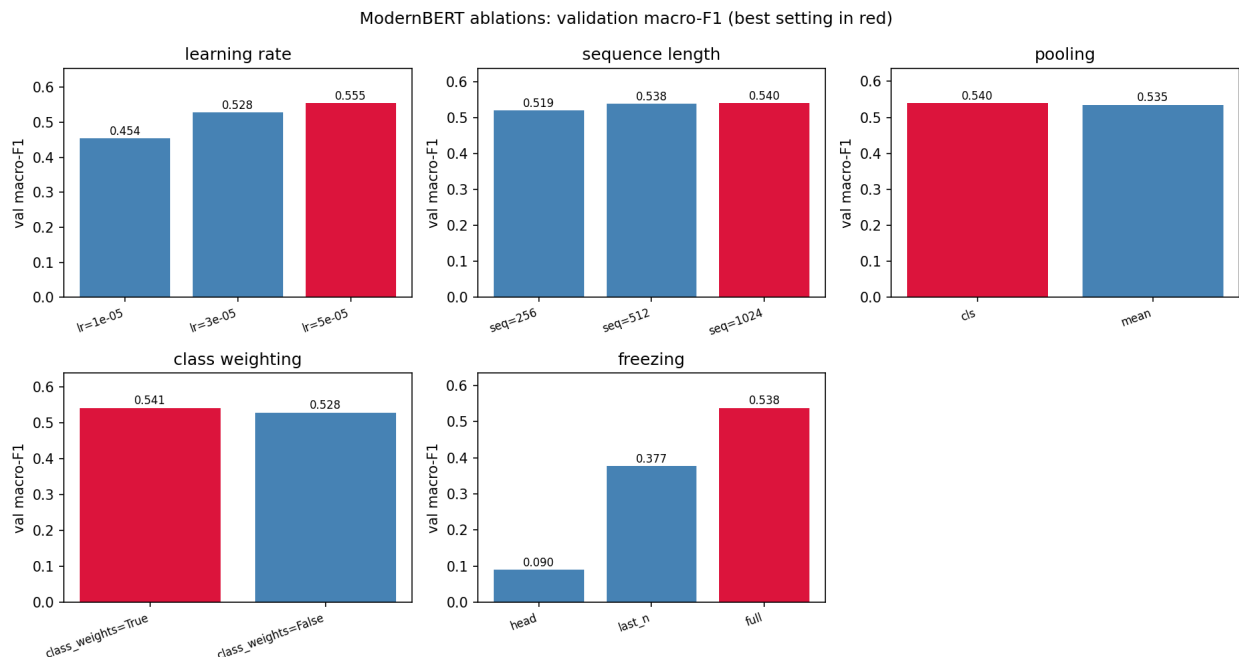


Figure 12: ModernBERT ablations across five axes, validation macro-F1, with the winning setting per axis highlighted.

Discussion

Two ceilings bound the absolute scores. The first is corpus size. With roughly 6,300 training rows spread across 25 classes, the average class has fewer than 300 examples, which limits what any model can learn and explains both the from-scratch model’s weakness and ModernBERT’s narrow margin over the baseline. The second is label noise. The dataset’s labels are single-source and unverified. Homepage2Vec showed that human-curated directory labels are systematically incomplete: crowd re-annotation of 807 sites raised the label count from 836 to 2,088, a factor of about 2.5 (Lugeon et al., 2022). Under such labels a well-calibrated model is penalized for predictions that are arguably correct but absent from the single recorded label, so the achievable macro-F1 is bounded below the value a clean-label task would permit.

The confusion matrices confirm that errors are not random but concentrate in semantically adjacent pairs such as Business and Career or Reference and Science, where the underlying content genuinely overlaps. This pattern argues for reading the confusion matrix and per-class scores together rather than relying on a single aggregate number, and it is consistent with the lexical overlap visible in the per-class vocabulary.

In the target deployment the classifier runs inside Dexar’s two-tier watcher, where a low-cost embedding similarity check handles the common case and a richer model resolves ambiguous samples. The content-based page classifier occupies the escalation tier for browser activity, the textual counterpart to the screenshot-based visual audit Dexar applies to non-browser windows. Operating inside the browser context also reduces the inference-time fetch limitation reported above, because the page has already been retrieved and rendered by the user’s browser, so the content is read from the active tab rather than fetched again over the network, which avoids the bot-wall and JavaScript-shell failure modes that affect a standalone fetch. The English-only constraint and the label-noise ceiling still apply and bound the reliability of the categorical signal, so a deployed verdict treats the predicted category as one input among several rather than a sole decision rule.

Problems Encountered

The English-only requirement removed about 38 percent of the data, far more than the proposal estimated, because the dataset is only about 62 percent English. This was the largest practical constraint and the

reason the final corpus is smaller than planned. Raw HTML frequently contained non-content pages such as Cloudflare bot walls, HTTP error pages, parked domains, and JavaScript-only shells, which required an explicit garbage detector to avoid training on boilerplate. Content longer than the sequence-length limit is truncated, which discards information on the longest pages. At inference time, live sites that block automated requests or render content only through JavaScript cannot be classified from their served HTML, and the pipeline surfaces these cases as explicit errors rather than guessing. Finally, the ablation showed that a frozen encoder collapses, which would have been a silent failure mode had fine-tuning depth not been tested directly.

Further Work

A zero-shot or few-shot large-language-model classifier, in the manner of Sasazawa and Sogawa (Sasazawa & Sogawa, 2025), would provide an additional baseline at low engineering cost. Relaxing the English-only constraint with a multilingual encoder would recover the discarded 38 percent of data and test cross-lingual transfer. Moving from single-label to multi-label classification would align the task with the genuinely multi-topic nature of many sites and with the Homepage2Vec formulation. Adding visual and HTML-metadata features alongside text, again following Homepage2Vec (Lugeon et al., 2022), could raise performance on classes that are visually distinctive but lexically ambiguous. Re-verifying a sample of labels would quantify the label-noise ceiling directly and put the reported macro-F1 in context.

Conclusion

The project delivers a complete pipeline from raw HTML to a live URL classifier, with a reproducible data preparation stage, three model families, and an ablation study. The fine-tuned ModernBERT reaches a test macro-F1 of 0.507 ± 0.004 over three seeds, ahead of a TF-IDF baseline at 0.473 and a from-scratch model at 0.319. The dominant lever is fine-tuning depth, and the absolute scores are bounded by a small English corpus and unverified labels. The result is a measured demonstration that a modern long-context encoder improves on classical and from-scratch baselines on this task, while the margin and the per-class errors reflect data constraints that the analysis identifies rather than hides.

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